An FSDS weight on the Q-index

Same table, a penalty on the action the stale policy would take

1 Where the weight sits

The control task already has a visit-bonus table $Q(s, a)$. The state s is the three-way sign of two known reference scores and the step. Putting the pseudo-outcome share into that state split the visits and raised the excess on the sign flip. The weight below does not add a coordinate.

Four reference policies are scored on every batch, on the same calendar as the scorecard race. The weight uses the two directional scores, the same two whose signs form the state. Write $w_{(1)}$ and $w_{(2)}$ for the larger and the smaller of their shares, and $a_{(1)}(s)$, $a_{(2)}(s)$ for the actions they prescribe. The penalty on an action is the share gap, and only on the action of the higher share, and only in states where the two actions differ:

$$\delta(s, a) = (w_{(1)} - w_{(2)})\, 1\{a = a_{(1)}(s)\}\, 1\{a_{(1)}(s) \neq a_{(2)}(s)\}.$$

The decision index is

$$Q(s, a) + c(s, a) - \eta\sigma \cdot \frac{H - h}{H} \cdot \delta(s, a),$$

with $\eta \in \{1, 2\}$ and σ the scale of the four reference returns on the first six rounds. The factor $(H - h)/H$ charges the episode-level penalty once along the horizon. Equal shares give $\delta = 0$, so the weight is inert while the two scores still carry the same residual. Agreement of the two actions also gives $\delta = 0$: the weight does not tax a button both scores would press.

The backup is unchanged. The target is the observed reward plus the max of the next raw table. The share is a statement about which reference score still explains the outcome. It is not a reward, so it does not enter the value being learned. An unvisited cell still carries the large visit bonus, so the weight cannot hide an action the table has not tried.

One variant turns the weight off on a round whose largest share belongs to the zero policy. That is the sign-flip symptom: the null score holds the mass, and a penalty aimed at a stale action has nothing stale to point at. Across pools, the fraction of post-shift rounds in which the zero policy holds the share has median 0.00 on relocation and 0.00 on the flip.

2 Relocation

The menu at $\eta = 2$ has median post excess 9.7. Visit-bonus Q with no weight is 15.9. The same table at $\eta = 1$ is 15.7, at $\eta = 2$ is 15.9, and at $\eta = 2$ with the zero-policy shut-off is 15.9. Discounted UCB on the menu is 15.3.

The $\eta = 2$ weight is lower than unweighted Q by more than the larger seed standard deviation on 0/6 pools. It is lower than the menu at $\eta = 2$ on 0/6, and the menu is lower than the weight on 2/6. Early excess is 12.5 without the weight and 12.5 with it. The first-action hit rate is 0.77 for the weight and 0.85 for the menu.

3 Sign flip

Unweighted visit-bonus Q is 7.4. The weight at $\eta = 1$ is 10.5, at $\eta = 2$ is 14.1, and with the zero-policy shut-off is 14.1. The menu at $\eta = 2$ is 132.0 and discounted UCB is 99.2.

The shut-off rule is lower than the always-on $\eta = 2$ weight by more than the seed standard deviation on 0/6 pools. Unweighted Q is lower than the always-on weight on 2/6 and lower than the shut-off rule on 2/6. The first-action hit rate is 0.64 without the weight, 0.54 with it always on, and 0.54 with the shut-off.

On relocation the $\eta = 2$ weight clears the unweighted table on 0/6 pools, and the menu at $\eta = 2$ stays lower on the pools where the table was already behind. On the sign flip the same weight moves the median from 7.4 to 14.1, and the first-action hit rate moves from 0.64 to 0.54. The zero-policy shut-off does not fire once the ramp has crossed one half. On the flip it fires on a few earlier rounds, and the post-shift excess stays within a point of the always-on weight (14.1 against 14.1). The share names which reference score has stopped explaining the outcome. Once a cell has a value, that name is not the button to press. The menu keeps the relocation index it already has. The visit-bonus table keeps the sign flip without this weight.

Table 1: Median, across six pools, of the four-seed mean post-shift excess return.

scenario	DUCB	UCB-H2	QUCB	QW1	QW2	QW2-Z
control_relocate	15.3	9.7	15.9	15.7	15.9	15.9
control_flip	99.2	132.0	7.4	10.5	14.1	14.1

Table 2: Relocation. Sum of per-round mean excess return. Seed standard deviation in parentheses.

pool	UCB-H2	QUCB	QW1	QW2	QW2-Z
AR(1)	9.8 (4.8)	10.0 (0.7)	9.8 (0.3)	10.0 (0.4)	10.0 (0.4)
California	3.3 (0.2)	12.3 (0.5)	11.1 (0.4)	11.7 (0.9)	11.7 (0.9)
covertype	7.2 (0.3)	24.0 (1.6)	25.7 (1.1)	26.3 (1.6)	26.3 (1.6)
Gaussian-12	9.5 (2.4)	9.6 (0.6)	9.8 (0.5)	10.4 (0.7)	10.4 (0.7)
Gaussian-30	20.4 (0.6)	19.4 (2.9)	20.5 (1.8)	20.1 (4.6)	20.1 (4.6)
Student-t_5	19.4 (1.3)	20.6 (3.3)	20.3 (3.8)	23.5 (4.7)	23.5 (4.7)

Table 3: Sign flip. Same excess. Seed standard deviation in parentheses.

pool	UCB-H2	QUCB	QW1	QW2	QW2-Z
AR(1)	88.5 (1.6)	2.5 (0.5)	2.6 (1.0)	2.8 (1.4)	2.7 (1.5)
California	78.2 (1.6)	2.5 (0.6)	2.3 (0.8)	3.0 (1.5)	3.0 (1.5)
covertype	160.9 (7.4)	21.1 (5.6)	26.8 (10.7)	28.5 (11.9)	28.5 (11.9)
Gaussian-12	105.5 (3.6)	3.1 (0.5)	2.6 (0.8)	2.6 (0.8)	2.6 (0.8)
Gaussian-30	160.4 (7.1)	12.6 (5.2)	18.4 (5.6)	25.2 (10.1)	25.2 (10.1)
Student-t_5	158.5 (4.9)	11.8 (7.1)	19.8 (5.5)	27.0 (9.3)	27.0 (9.3)

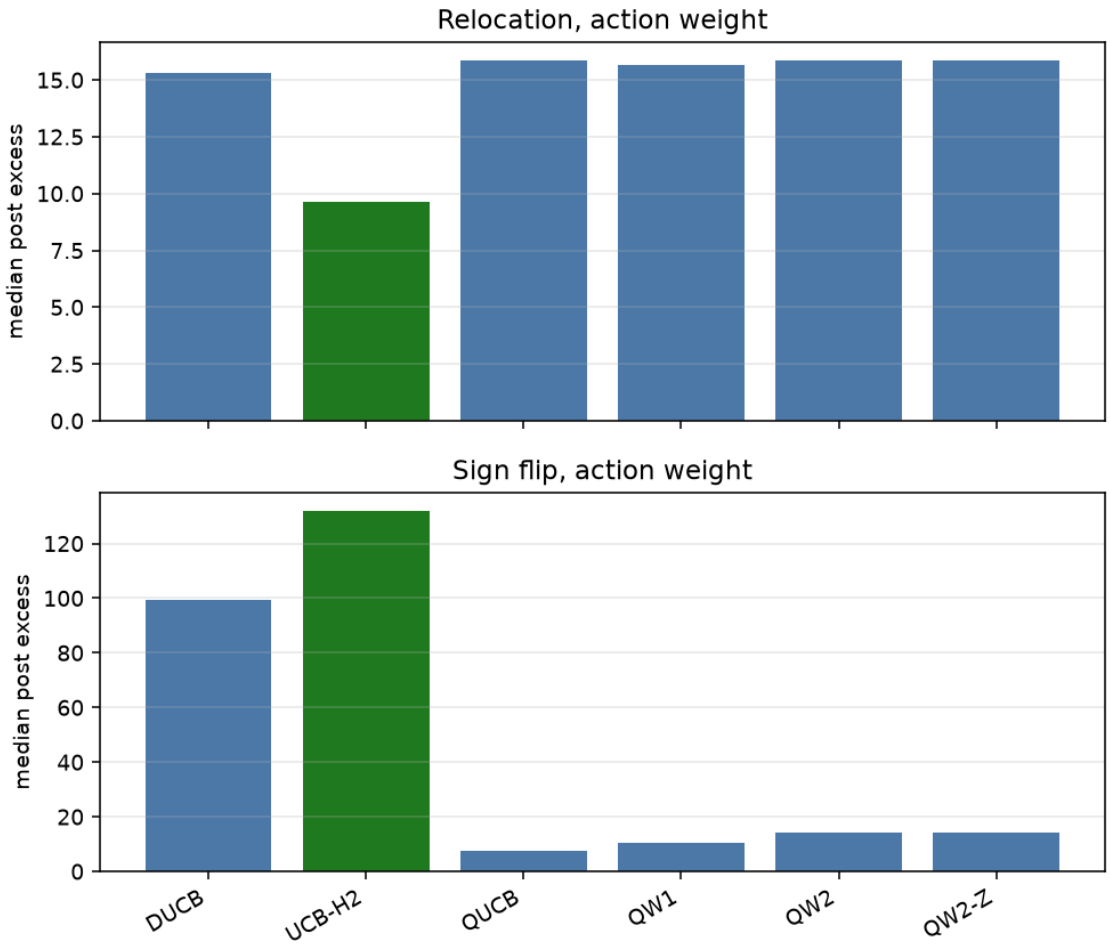


Figure 1: Median post-shift excess. The menu at $\eta = 2$ is marked in green. QW1 and QW2 put the share on the action index. QW2-Z turns that weight off when the zero policy holds the share.